

5XTCO - June 2020

Q1

a) $z_L = (15 + j39) \Omega$

$z_1 = 150 \Omega$

$$\frac{z_L}{z_1} = \frac{15}{150} + j \frac{39}{150} = 0.1 + j 0.26 \rightarrow \text{Find in SC}$$

$$|P_L| = 0.83$$

$$\angle = 151^\circ$$

b) $l_1 = 0.321 \text{ m}$, $\lambda = \frac{c}{f} = \frac{3 \cdot 10^8 \text{ m/s}}{400 \cdot 10^6 \text{ Hz}} = 0.75 \text{ m}$

$$\frac{l_1}{\lambda} = 0.428 \Leftrightarrow l_1 = 0.428 \lambda \rightarrow \text{Turn } 0.428 \lambda \text{ towards generator}$$

$$\frac{z_L'}{z_1} = 0.1 - j 0.2 \leftarrow \text{in SC}$$

$$\underline{\underline{z_L' = (15 - j30) \Omega}}$$

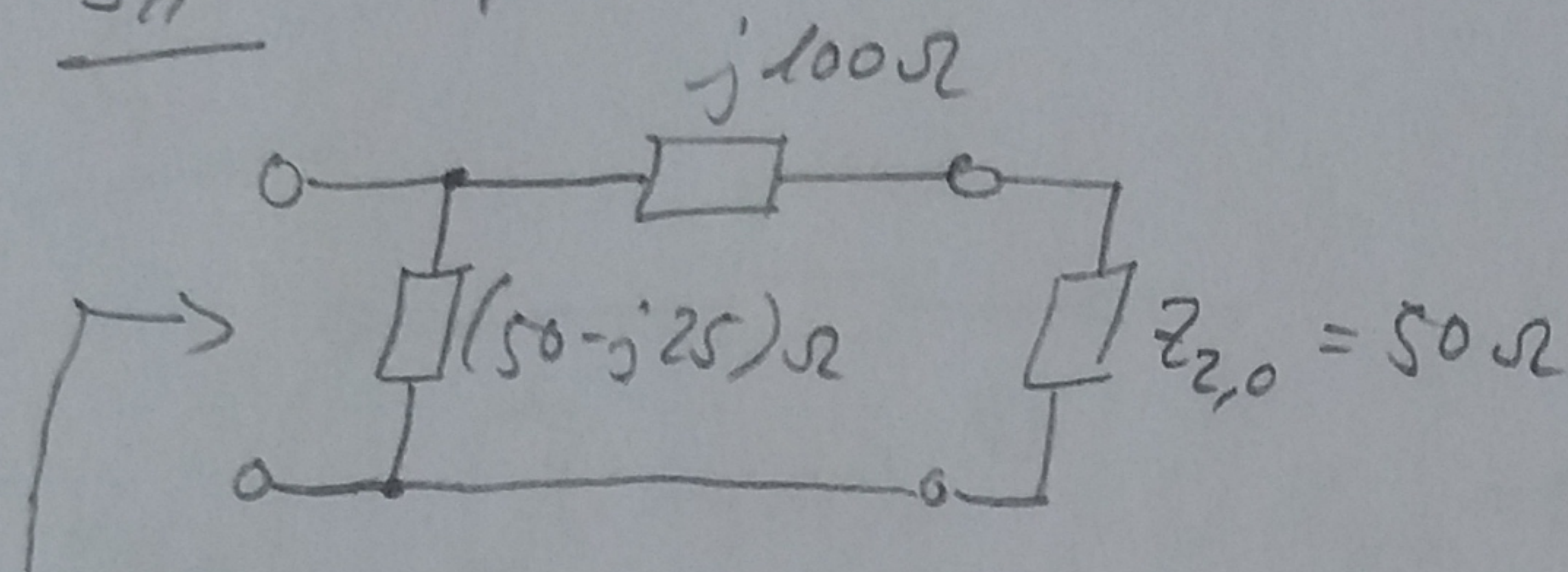
c) $\frac{z_L'}{z_0} = \frac{15}{45} - j \frac{30}{45} = 0.2 - j 0.4 \rightarrow \text{Find in SC}$

$$\text{From SC: } \frac{Y_L'}{Y_0} = 1 + j 2 \Rightarrow \underline{\underline{z_s = 0 \Omega}}$$

$$Y_p = (-j 2) Y_0 = -j \frac{2}{45} \frac{1}{\Omega}$$

$$\underline{\underline{z_p = j 27.5 \Omega}}$$

d) S_{11} : Ref. impedance $z_0 = 50\Omega$



$$\Gamma_1 = S_{11}$$

$$Z_{in,1} = \underbrace{(50 - j25)\Omega}_{\text{find in SC}} \parallel \underbrace{(50 + j100)\Omega}_{\text{find in SC}} \rightarrow Z_2 = \frac{50}{50} + j \frac{100}{50} = \frac{1 + j2}{Z_0}$$

$$\frac{Z_1}{Z_0} = \frac{50}{50} - j \frac{25}{50}$$

$$\frac{Z_1}{Z_0} = 1 - j0.5$$

$$\frac{Y_1}{Y_0} = 0.8 + j0.4$$

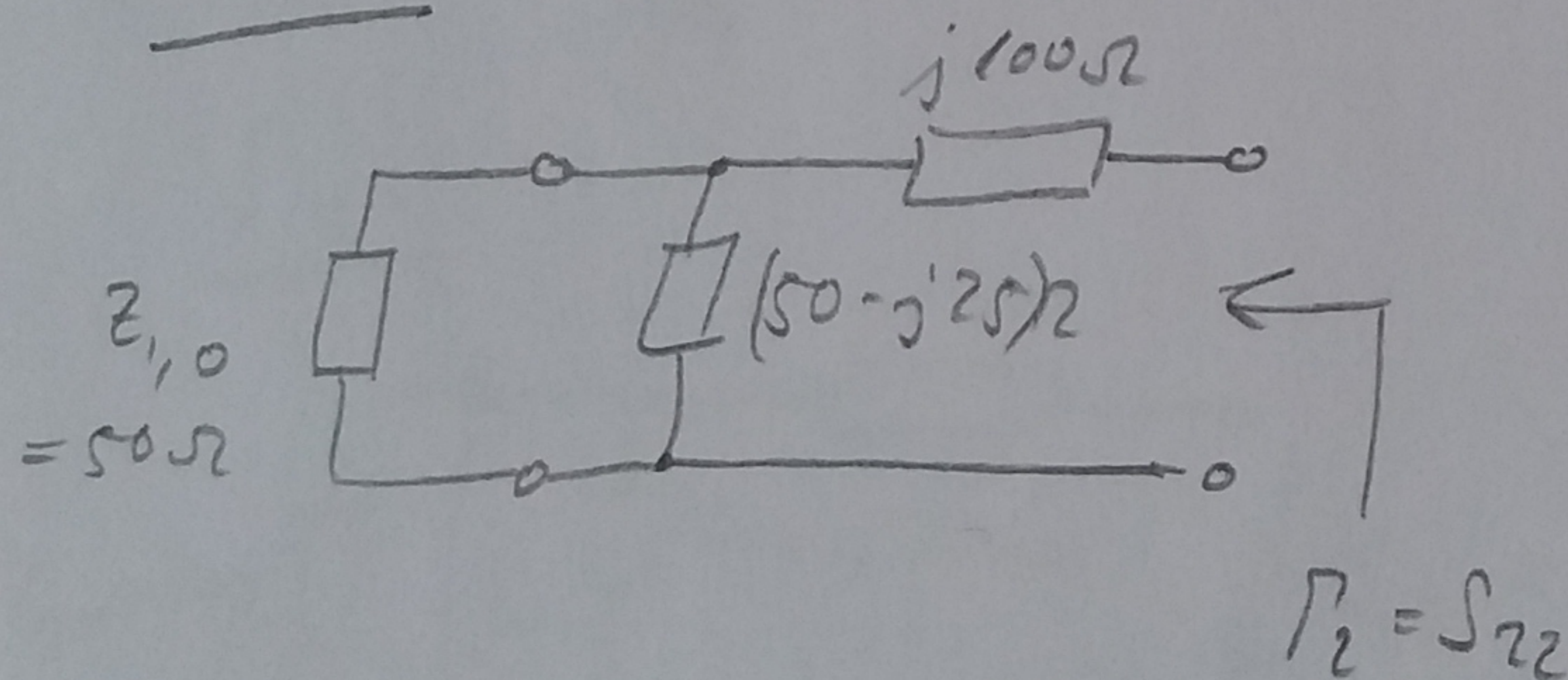
$$\frac{Y_2}{Y_0} = 0.2 - j0.4$$

$$\boxed{\frac{Y_1 + Y_2}{Y_0} = 1} \text{ center of SC!}$$

$$\Rightarrow Z_{in,1} = 50\Omega$$

$$\Rightarrow \underline{\underline{S_{11} = 0}}$$

S_{22} :



$$Z_{in,2} = j100\Omega + \left\{ 50\Omega \parallel (50 - j25)\Omega \right\}$$

$$\frac{Y_1}{Y_0} = 1 + 0.8 + j0.4$$

$$\frac{Y_1}{Y_0} = 1.8 + j0.4$$

$$\frac{Z_1}{Z_0} = 0.53 - j0.12$$

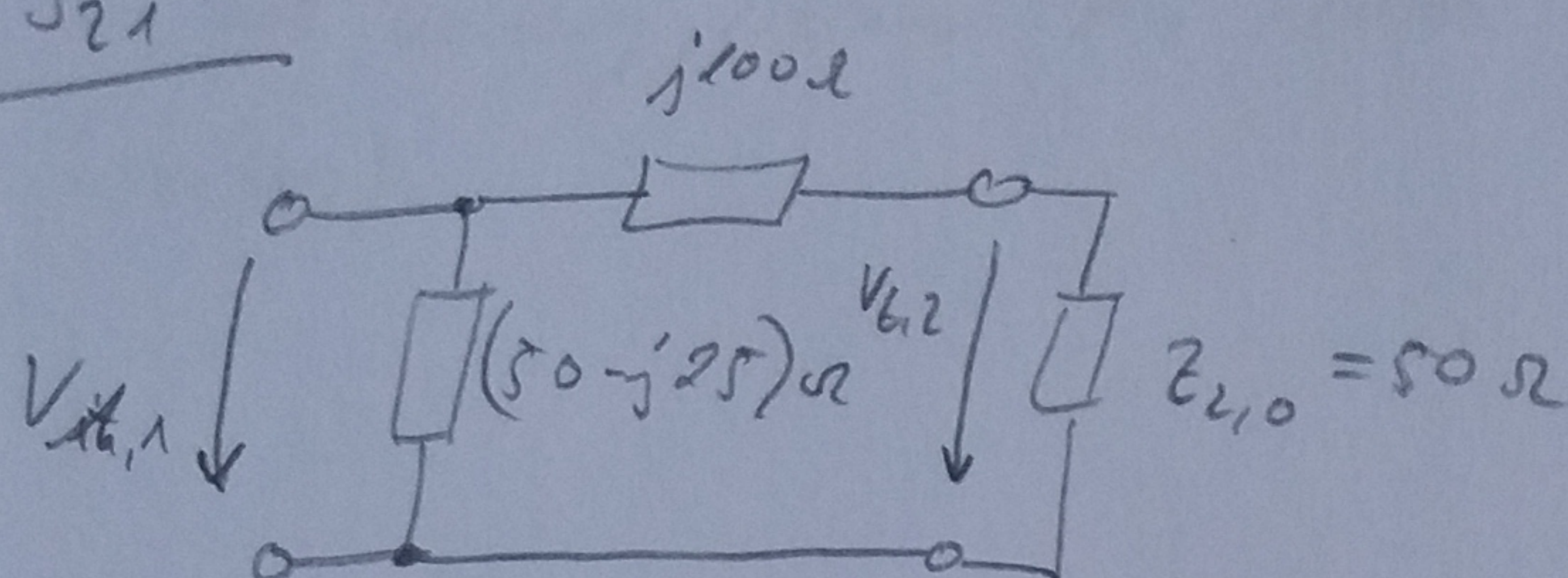
$$\cancel{j100\Omega} + Z_1 = 26.5 - j6$$

$$Z_{in,2} = (26.5 - j9.4)\Omega$$

$$\underline{\underline{S_{22} = \frac{Z_{in,2} - Z_0}{Z_{in,2} + Z_0} = 0.48 + j0.64}}}$$

d) cont'd

S_{21} :



$V_{t,1} = (1 + S_{11}) V_{in,1}$; $V_{in,1}$: incident voltage wave on port 1 (from the left)

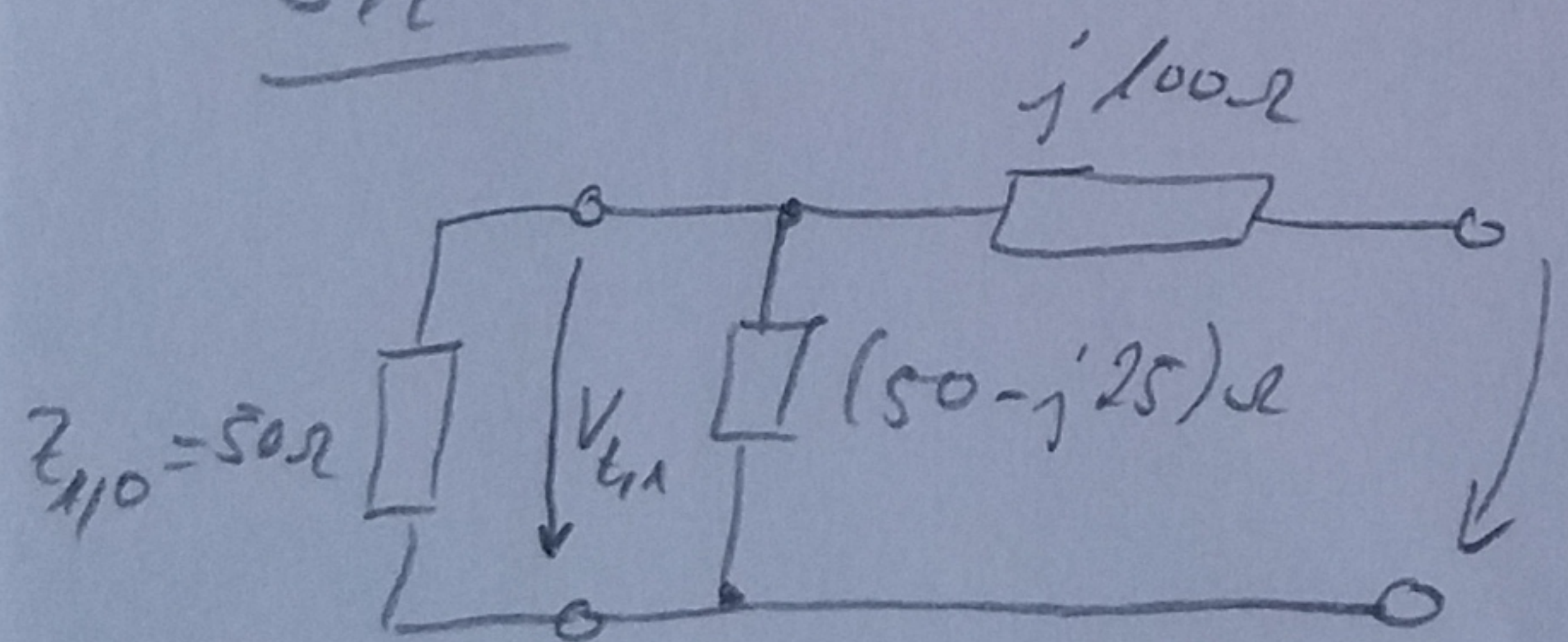
$$V_{t,2} = \frac{Z_{L,0}}{Z_{L,0} + j100\Omega} V_{t,1} = \frac{Z_{L,0}}{Z_{L,0} + j100\Omega} (1 + S_{11}) V_{in,1}; \quad \frac{V_{t,2}}{\text{power in } Z_{L,0} \text{ (i.e. port 2)}}$$

$$= \frac{50\Omega}{(50 + j100)\Omega} (1 + 0) V_{in,1}$$

$$= \frac{1}{1 + j2} V_{in,1} = \frac{1 - j2}{5} V_{in,1} = (0.2 - j0.4) V_{in,1}$$

$$\underline{\underline{S_{21}}} = \underbrace{\frac{V_{t,2}}{\sqrt{Z_0}}}_{b_2} \cdot \underbrace{\frac{\sqrt{Z_0}}{V_{in,1}}}_{\frac{1}{a_1}} = \frac{V_{t,2}}{V_{in,1}} = \underline{\underline{0.2 - j0.4}}$$

S_{12} :



$$V_{t,2} = (1 + S_{22}) V_{in,2}$$

incident voltage wave on port 2 (from the right)

$V_{t,1}$: accepted/dissipated power in $Z_{L,0}$ (i.e. port 1)

$$V_{t,1} = \frac{50\Omega \parallel (50 - j25)\Omega}{j100 + \{50\Omega \parallel (50 - j25)\Omega\}} (1 + S_{22}) V_{in,2}$$

$$= \frac{26.5 - j6}{j100 + 26.5 - j6} (1 + S_{22}) V_{in,2}$$

$$= \frac{26.5 - j6}{26.5 + j94} (1 + S_{22}) V_{in,2}$$

$$= 0.48 + j0.64$$

$$\underline{\underline{S_{12}}} = \frac{V_{t,1}}{V_{in,2}} = \underline{\underline{0.2 - j0.4}} = S_{21}$$

! This result could have been known without calculations. The network is passive and must be reciprocal $\Rightarrow$ S-matrix must be symmetrical!

c) The scattering matrix of the setup shown in Fig. 2 is not unitary, because the network contains lossy components (resistors).